

Lesson 3-3 Remediation

Polynomial Identities & Binomial Theorem

Name: _____ Period: _____ Date: _____

Part 1: Finding Cube Roots

Warm-Up

To use the Sum or Difference of Cubes identity, you first need to figure out what a and b are. That means finding the **cube root** — what number, when cubed, gives you each term?

(a) $8 = (\quad)^3$ so the cube root of 8 is _____

(b) $27 = (\quad)^3$ so the cube root of 27 is _____

(c) $64 = (\quad)^3$ so the cube root of 64 is _____

(d) $125 = (\quad)^3$ so the cube root of 125 is _____

Now try these — the cube root can include a **variable**:

(e) $125x^3 = (\quad)^3$ cube root: _____

(f) $8x^3 = (\quad)^3$ cube root: _____

(g) $27x^6 = (\quad)^3$ cube root: _____

(h) $64x^9 = (\quad)^3$ cube root: _____

Sum of Cubes: $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$

Difference of Cubes: $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

Problem 1: Factor $27 + 8x^3$

Step 1: Is this a sum or difference? Which identity applies?

Step 2: Find a and b .

$$a = \underline{\hspace{2cm}} \quad \text{because} \quad \left(\underline{\hspace{1cm}} \right)^3 = 27$$

$$b = \underline{\hspace{2cm}} \quad \text{because} \quad \left(\underline{\hspace{1cm}} \right)^3 = 8x^3$$

Step 3: Substitute a and b into the identity. Replace *every* a and *every* b :

$$\left(\underbrace{\hspace{1.5cm}}_a + \underbrace{\hspace{1.5cm}}_b \right) \left(\underbrace{\hspace{1.5cm}}_{a^2} - \underbrace{\hspace{1.5cm}}_{a \cdot b} + \underbrace{\hspace{1.5cm}}_{b^2} \right)$$

Step 4: Now simplify each piece. Show your arithmetic:

$$a^2 = \left(\underline{\hspace{1cm}} \right)^2 = \underline{\hspace{2cm}}$$

$$a \cdot b = \left(\underline{\hspace{1cm}} \right) \left(\underline{\hspace{1cm}} \right) = \underline{\hspace{2cm}}$$

$$b^2 = \left(\underline{\hspace{1cm}} \right)^2 = \underline{\hspace{2cm}}$$

Final factored form:

$$27 + 8x^3 = \left(\underline{\hspace{2cm}} \right) \left(\underline{\hspace{2cm}} \right)$$

Problem 2: Factor $64x^3 - 125$

Step 1: Is this a sum or difference? Which identity applies?

Step 2: Find a and b .

$$a = \underline{\hspace{2cm}} \quad \text{because} \quad \left(\underline{\hspace{2cm}} \right)^3 = 64x^3$$

$$b = \underline{\hspace{2cm}} \quad \text{because} \quad \left(\underline{\hspace{2cm}} \right)^3 = 125$$

Step 3: Substitute into the identity:

$$\left(\underbrace{\hspace{1.5cm}}_a - \underbrace{\hspace{1.5cm}}_b \right) \left(\underbrace{\hspace{1.5cm}}_{a^2} + \underbrace{\hspace{1.5cm}}_{a \cdot b} + \underbrace{\hspace{1.5cm}}_{b^2} \right)$$

Step 4: Simplify each piece:

$$a^2 = \left(\underline{\hspace{1cm}} \right)^2 = \underline{\hspace{2cm}}$$

$$a \cdot b = \left(\underline{\hspace{1cm}} \right) \left(\underline{\hspace{1cm}} \right) = \underline{\hspace{2cm}}$$

$$b^2 = \left(\underline{\hspace{1cm}} \right)^2 = \underline{\hspace{2cm}}$$

Final factored form:

$$64x^3 - 125 = \left(\underline{\hspace{2cm}} \right) \left(\underline{\hspace{2cm}} \right)$$

Problem 3: Factor $x^3 + 216$

(On your own — show all 4 steps)

Problem 4: Factor $1000 - 27x^3$

(On your own — show all 4 steps)

Part 3: Powers of Negative Numbers

The Step Most Students Skip

Before you can expand something like $(x - 3)^n$, you need to know the powers of (-3) .
 Fill in the table below by **multiplying one step at a time**. Watch the signs carefully!

Power	Multiply it out	Value	+ or - ?
$(-3)^0$	anything to the 0 is 1	1	+
$(-3)^1$	-3	-3	-
$(-3)^2$	$(-3)(-3) =$		
$(-3)^3$	$(-3)(-3)(-3) = (\quad)(-3) =$		
$(-3)^4$	$(\text{previous answer}) \times (-3) =$		
$(-3)^5$	$(\text{previous answer}) \times (-3) =$		

Pattern: Odd powers of a negative number are always _____.

Even powers of a negative number are always _____.

Part 4: Binomial Expansion — One Term at a Time

Guided Practice

Pascal's Triangle coefficients for $n = 4$: 1 4 6 4 1

The setup: $(x - 3)^4 = 1 \cdot x^4 \cdot (-3)^0 + 4 \cdot x^3 \cdot (-3)^1 + 6 \cdot x^2 \cdot (-3)^2 + 4 \cdot x^1 \cdot (-3)^3 + 1 \cdot (-3)^4$

Simplify each term. Use your table from Part 3!

Term 1: $1 \cdot x^4 \cdot (-3)^0 = 1 \cdot x^4 \cdot 1 = x^4$ *(done for you)*

Term 2: $4 \cdot x^3 \cdot (-3)^1 = 4 \cdot x^3 \cdot (\quad) = \underline{\hspace{2cm}}$

Term 3: $6 \cdot x^2 \cdot (-3)^2 = 6 \cdot x^2 \cdot (\quad) = \underline{\hspace{2cm}}$

Term 4: $4 \cdot x^1 \cdot (-3)^3 = 4 \cdot x \cdot (\quad) = \underline{\hspace{2cm}}$

Term 5: $1 \cdot (-3)^4 = 1 \cdot (\quad) = \underline{\hspace{2cm}}$

Final Answer:

$(x - 3)^4 = \underline{\hspace{4cm}}$

Check your signs: Do the signs alternate $+, -, +, -, +$? Why does that happen?

Now you try: Expand $(x - 2)^4$

Pascal's Triangle coefficients for $n = 4$: 1 4 6 4 1

First: Build your powers table for (-2) .

Power	Work	Value
$(-2)^0$		
$(-2)^1$		
$(-2)^2$		
$(-2)^3$		
$(-2)^4$		

Then: Simplify each term.

Term 1: $1 \cdot x^4 \cdot (-2)^0 =$ _____

Term 2: $4 \cdot x^3 \cdot (-2)^1 =$ _____

Term 3: $6 \cdot x^2 \cdot (-2)^2 =$ _____

Term 4: $4 \cdot x^1 \cdot (-2)^3 =$ _____

Term 5: $1 \cdot (-2)^4 =$ _____

Final Answer:

$(x - 2)^4 =$ _____

Check your signs: Do they alternate? _____ Does your last term have no x ? _____